WON KO

https://won-ko.github.io

SUMMARY

Senior Researcher in autonomous robotics specializing in multi-modal perception and sensor fusion (camera/LiDAR/IMU), sensor calibration, 3D SLAM, and motion planning/control for reliable real-world autonomy.

EDUCATION

University of California, Santa Cruz — August 2022
Master of Science in Computer Engineering
Emphasise in Robotics and Control
MS, "Autonomously Driving Tricycle with Monocular Vision"
Advisor: Prof. Dejan Milutinović

University of California, Santa Cruz — December 2019
Bachelor of Science in Computer Engineering
Concentration on Robotics and Control

EXPERIENCE

Senior Researcher / Future Tech R&D Lab, Mobyus — November 2024 - Present

- Research and develop autonomy and perception for AMR/AFL platforms: multi-modal sensor fusion, state estimation (3D SLAM), and motion planning/control on real hardware.
- Developed and validated 3D SLAM pipelines (LiDAR–inertial with ESKF; visual–inertial/VIO–VSLAM); achieved $\leq$30 mm localization error on testbeds and improved repeatability in field trials.
- Built multi-sensor perception for obstacle detection and scene understanding (camera + LiDAR fusion) to support safe navigation in dynamic environments.
- Designed sensor calibration algorithms using least-squares optimization: LiDAR$\leftrightarrow$IMU and camera$\leftrightarrow$IMU extrinsic calibration; joint calibration across LiDAR/IMU/camera.
- Productionized and deployed autonomy modules with reliability features (startup checks, recovery behaviors, E-stop/watchdogs), PLC workflow integration, and operational logging/telemetry; maintained build/release workflows (colcon, containers, CI/CD).

Researcher / Robotics and Control Lab, UC Santa Cruz — September 2020 - August 2023

- Researched stochastic nonlinear control and feedback and control systems, embedded systems, sensor fusion, guidance, navigation, and control (GNC) system identification, especially GPS-free navigation and mobile robot autonomy.
- Implemented estimation/control prototypes and evaluated robustness under real sensor noise and timing constraints.

Teaching Assistant / University of California, Santa Cruz — Spring 2022, Fall 2021
Mechatronics (ECE 118)
Santa Cruz, CA

- Led lab sections on event-driven programming, hierarchical state machines (HSMs), sensor interfacing, and end-to-end system integration; provided debugging support and graded coursework with the TA team.
- Maintained and repaired lab hardware; performed safety checks and managed spare parts/inventory to minimize downtime.

Teaching Assistant / University of California, Santa Cruz Spring 2021, Winter 2022
Computer Architecture (CSE 120) *Santa Cruz, CA*

- Led RISC-V and hardware design labs using Verilog/SystemVerilog; guided students through simulation, debugging, and design verification.
- Held weekly office hours and graded assignments/exams; supported performance analysis and correctness debugging for lab implementations.

PROJECTS

Autonomous Forklifts (AFL) November 2024 - October 2025

- Developed Frenet-frame optimal-trajectory planning with B-spline smoothing for obstacle avoidance and precise docking; implemented a three-wheeled motion controller for AFL platforms (SD-type motors).
- Built a 3D localization pipeline integrating LiDAR–IMU (ESKF-based LIO); tuned for stable pose and repeatable navigation in cluttered environments; validated on real hardware (e.g., $\leq$30 mm position error on testbeds).
- Designed multi-sensor obstacle detection using camera + LiDAR fusion (RealSense D435f depth + 2D SICK LiDAR) to improve robustness under occlusion and reflective surfaces.
- Developed pallet perception using SICK Visionary-T Mini: estimated pallet size and fork-pocket (hole) centers with a Transformer-based model and CAD-based geometry; achieved hole detection at 4 m with $\sim$0.5° angular error for precise insertion/alignment.
- Built truck/trailer free-space perception using ZED2i stereo with object detection models (YOLOv12, RF-DETR) to estimate available loading space and placement feasibility.

Continental Intelligent Intersection – *co-sponsored by Continental AG* September 2018 - June 2019

- Developed and evaluated ML models (CNN, RNN, SVM, Hidden Markov Model) for traffic-scene understanding and risk prediction at intersections.
- Customized YOLOv3 for project-specific object detection and trained on curated datasets from heterogeneous sensors (short/long-range radar and camera).
- Built a data pipeline for sensor synchronization/labeling and deployed inference to detect safety-critical scenarios (e.g., pedestrian crossings, left-turn conflicts) using Python + OpenCV.

Autonomous Robot April 2019 - June 2019

- Built an autonomous mobile robot using Uno32, servos, DC motors, and custom sensor modules to navigate a standardized field and execute goal-directed behaviors under constraints.
- Designed a spring-based shooting mechanism and created electrical test harnesses to validate subsystems before full integration.
- Implemented a hierarchical state machine (HSM) in Embedded C for robust event-driven control.
- Calibrated and validated sensors (tape sensor, beacon detector, track wire) and iterated on control/behavior logic for repeatable performance.

TECHNICAL SKILLS

Languages	Embedded C/ C++, Python (OpenCV, TensorFlow, Pytouch), Verilog/System Verilog.
Software	Isaac SIM, Git, Docker, MATLAB/Simulink, V-REP, Solidworks, MPLab, Xilinx.
Operating Systems	ROS, Unix/Linux system (FreeBSD, Ubuntu, Debian), Windows